Lab Manual - LAB 3

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LAB 3: IP Addressing and IPv4 Fundamentals

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IP Addressing and IPv4 Fundamentals

Learning Objectives

By the end of this lab, students should be able to:

1. Define an IP address and explain its purpose.
2. Identify the structure of an IPv4 address.
3. Explain bits, bytes, and octets.
4. Convert IPv4 octets between decimal and binary.
5. Explain the purpose of a subnet mask.
6. Identify network and host portions of an IPv4 address.
7. Explain the role of a default gateway.
8. Determine whether two IPv4 addresses belong to the same /24 network.

What is an IP Address?

An **IP address (Internet Protocol address)** is a logical address assigned to a device on a network. It allows devices to:

- Identify one another
- Communicate with one another
- Send and receive data
- Determine where data should be delivered

For example, 192.168.10.5 can be an IPv4 address assigned to a computer. Think of an IP address like a **digital address for a device**.

Real-world analogy: Imagine sending a letter.

You need: **House Address → Street → City → Country**

Similarly, when data is sent across a network, devices need logical addressing information to determine: **Source → Destination**

Why Do We Need IP Addresses?

Consider a network containing three computers:

- PC 1
- PC 2
- PC 3

If all computers simply existed without addresses, the network would have no logical way to identify the intended destination. With IP addresses:

- PC 1 → 192.168.10.1
- PC 2 → 192.168.10.2
- PC 3 → 192.168.10.3

Now each device has a logical identity. For example:

PC 1: 192.168.10.1
|
| Send data to
↓
PC 2: 192.168.10.2

The destination IP tells the network where the data needs to go.

IPv4

IPv4 (Internet Protocol version 4) is one of the most widely used versions of the Internet Protocol. An IPv4 address contains **32 bits.** These 32 bits are divided into **four groups of 8 bits**.

32 bits
├── 8 bits
├── 8 bits
├── 8 bits
└── 8 bits

Each group of 8 bits is called an **octet**.

Therefore:

IPv4 Address → 4 Octets → 8 + 8 + 8 + 8 → 32 bits

Example:

192.168.10.5

It contains four octets:

192	168	10	5
↓	↓	↓	↓
Octet 1	Octet 2	Octet 3	Octet 4

Bits, Bytes, and Octets

Before converting IPv4 addresses, students need to understand three basic terms.

Bit:

A **bit** is the smallest unit of digital information. It can have only two values 0 or 1.

Byte:

A **byte** contains 8 bits.

1 Byte = 8 bits

For example, 11001010 is one byte.

Octet:

In IPv4, each group of 8 bits is called an **octet**. Therefore:

1 IPv4 address
= 4 octets
= 32 bits

IPv4 Dotted Decimal Notation

IPv4 addresses are normally written in dotted decimal notation. Example:

192.168.10.25

The dots separate the four octets.

192	168	10	25
↓	↓	↓	↓
Octet 1	Octet 2	Octet 3	Octet 4

Each octet can have a value from 0 to 255. Therefore, a valid IPv4 address looks like:

0–255 . 0–255 . 0–255 . 0–255

Examples:

- 192.168.1.10
- 10.0.0.25
- 172.16.5.100

Why is the Maximum Octet Value 255?

Each octet contains 8 bits. The binary place values are:

Bit	128	64	32	16	8	4	2	1
Position	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

If all 8 bits are 1:

11111111

The decimal value is:

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1
= 255

Therefore:

- Minimum = 00000000 = 0
- Maximum = 11111111 = 255

Binary Place Values

Memorise the following table:

128	64	32	16	8	4	2	1

These values are used to convert between binary and decimal.

Example

Convert: 11000000 to decimal.

Write the place values:

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

Add the values where the bit is 1:

128 + 64 = 192

Therefore:

11000000 = 192

Decimal-to-Binary Conversion

To convert a decimal number into binary, determine which place values add up to that number.

Example 1: Convert 10 to binary

Place values:

128	64	32	16	8	4	2	1

We need:

10 = 8 + 2

Therefore:

0	0	0	0	1	0	1	0

So, 10 = 00001010

2nd Approach:

2	10	
2	5	0
2	2	1
	1	0

So, 10 = 1010

Or

10 = 00001010

Example 2: Convert 25 to binary

25 = 16 + 8 + 1

Therefore:

128	64	32	16	8	4	2	1
0	0	0	1	1	0	0	1

So, 25 = 00011001

Example 3: Convert 192 to binary

192 = 128 + 64

Therefore:

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

So, 192 = 11000000

Binary-to-Decimal Conversion

To convert binary to decimal, add the place values corresponding to 1.

Convert 10101000 to Decimal:

Place values:

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

Therefore:

128 + 32 + 8 = 168

So,

10101000 = 168

IPv4 Address in Binary

An entire IPv4 address contains four octets.

Example:

192.168.1.10

Convert each octet:

192 = 11000000
168 = 10101000
1 = 00000001
10 = 00001010

Therefore:

192.168.1.10

becomes:

11000000.10101000.00000001.00001010

This contains:

8 + 8 + 8 + 8 = 32 bits

Network Portion and Host Portion

An IPv4 address has two logical parts:

Network Portion: Identifies the network to which the device belongs.

Host Portion: Identifies the individual device within that network.

For example:

192.168.10.25

The exact division between network and host depends on the **subnet mask**. For a /24 network:

IP Address:
192.168.10.25

Network:
192.168.10.0

Host:
25

Conceptually:

192.168.10 | 25
Network | Host

Subnet Mask

A **subnet mask** determines which portion of an IPv4 address represents the network and which portion represents the host. A common subnet mask is:

255.255.255.0

This is also written as /24. The /24 means that the first **24 bits** belong to the network portion.

/24

11111111.11111111.11111111.00000000

Therefore:

Network bits = 24
Host bits = 8

Example: IP Address and Subnet Mask

Consider:

IP Address: 192.168.10.25
Subnet Mask: 255.255.255.0

The /24 mask means:

192.168.10 | 25
Network | Host

Therefore:

Network Address = 192.168.10.0
Host Address = 25

For a basic /24 network, the usable host range is:

192.168.10.1 – 192.168.10.254

The network address is:

192.168.10.0

The broadcast address is:

192.168.10.255

Default Gateway

A **default gateway** is the device that provides a path from one network to another. Usually, the default gateway is a **router interface**.

Example:

PC0: 192.168.10.10
|
|

Switch
|
|
Router
192.168.10.1
|
|
192.168.20.1

If PC0 wants to communicate with a device on another network, it sends the traffic toward its default gateway. For example:

PC0: 192.168.10.10
|
↓
Default Gateway
192.168.10.1
|
↓
Other Network
192.168.20.0/24

Important

Devices communicating within the same local network generally do **not** need the default gateway for that local communication. The gateway becomes important when communication needs to leave the local network.

Same Network or Different Network?

Consider:

PC0: 192.168.10.10
PC1: 192.168.10.20

Subnet Mask:
255.255.255.0

Both belong to:

192.168.10.0/24

Therefore, they are in the **same network**.

Now consider:

PC0: 192.168.10.10
PC1: 192.168.20.20

Subnet Mask:
255.255.255.0

Their networks are:

PC0 → 192.168.10.0/24
PC1 → 192.168.20.0/24

They belong to **different networks**.

Key Takeaways

Remember these important points:

- IPv4 → 32 bits → 4 octets → 8 bits per octet
- 1 Byte = 8 bits
- IPv4 octet range = 0–255
- 255.255.255.0 = /24
- IP Address + Subnet Mask → Network Identity
- Same Network → Local communication
- Different Network → Router